



UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

CSA3023

WEB-BASED APPLICATION DEVELOPMENT

**LAB 1:
DEVELOPMENT ENVIRONMENT SETUP FOR JAVA
WEB APPLICATIONS**

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Prepared for:
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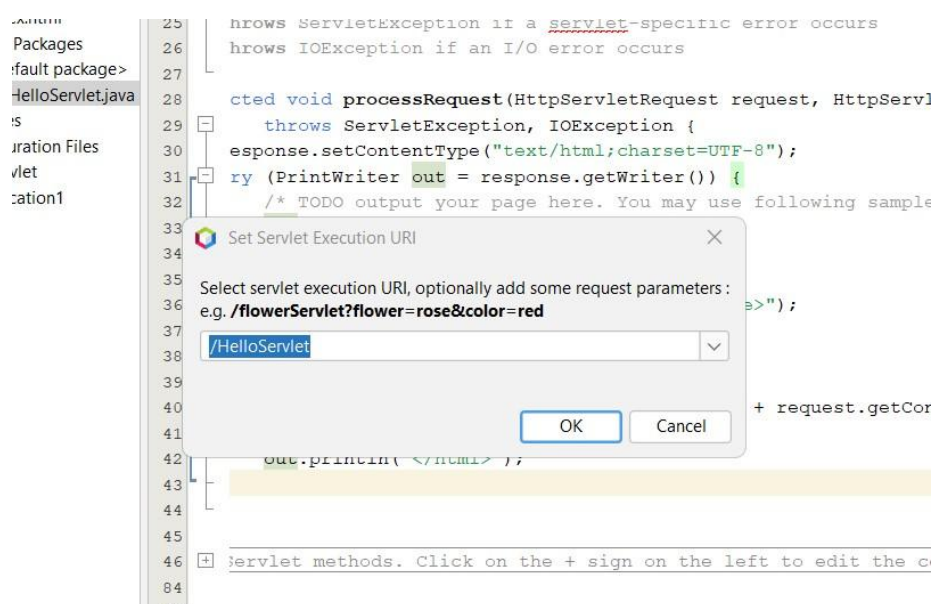
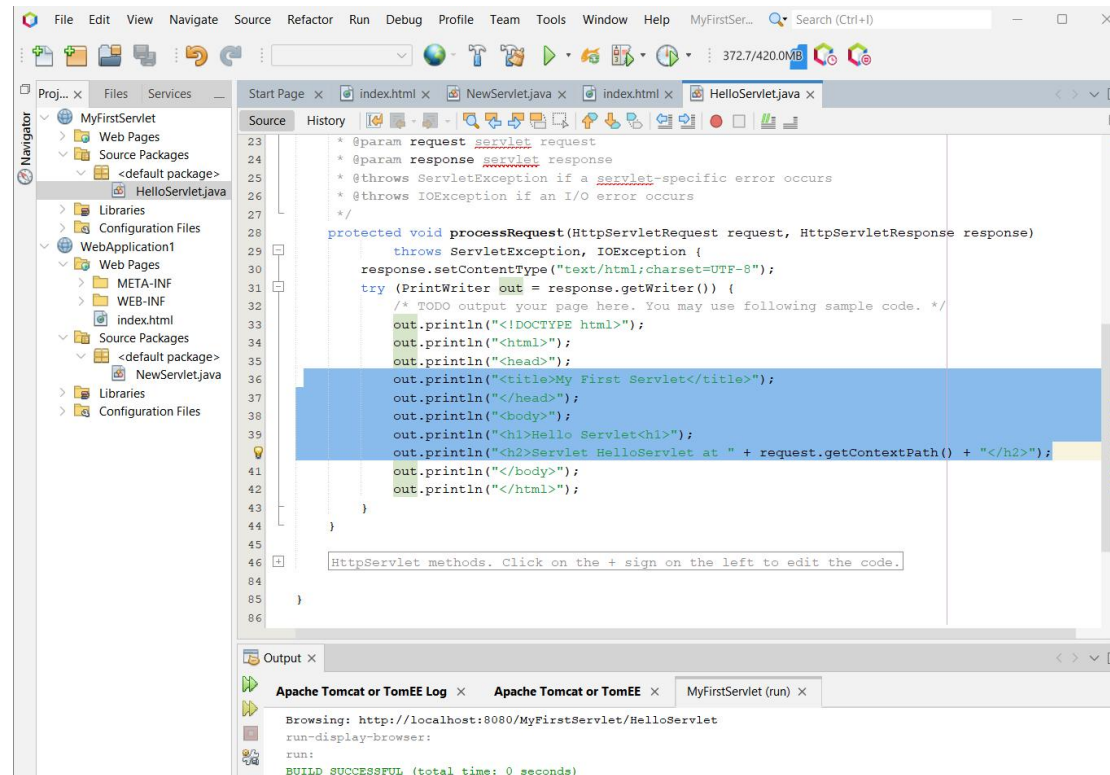
Date:
1 APRIL 2026

TASK 6: WRITING A SIMPLE JAVA SERVLET

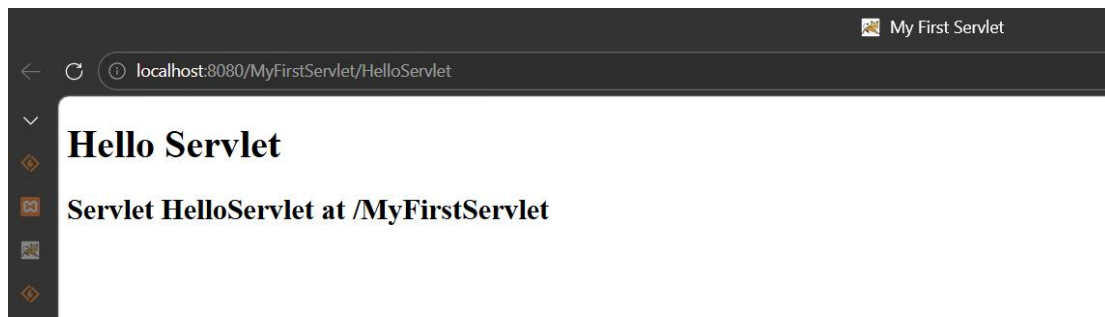
Objective: Develop and execute a simple Java Servlet using Apache NetBeans and Apache Tomcat.

1. Modify the existing code as instructed (refer to the provided figure).

ANSWER:



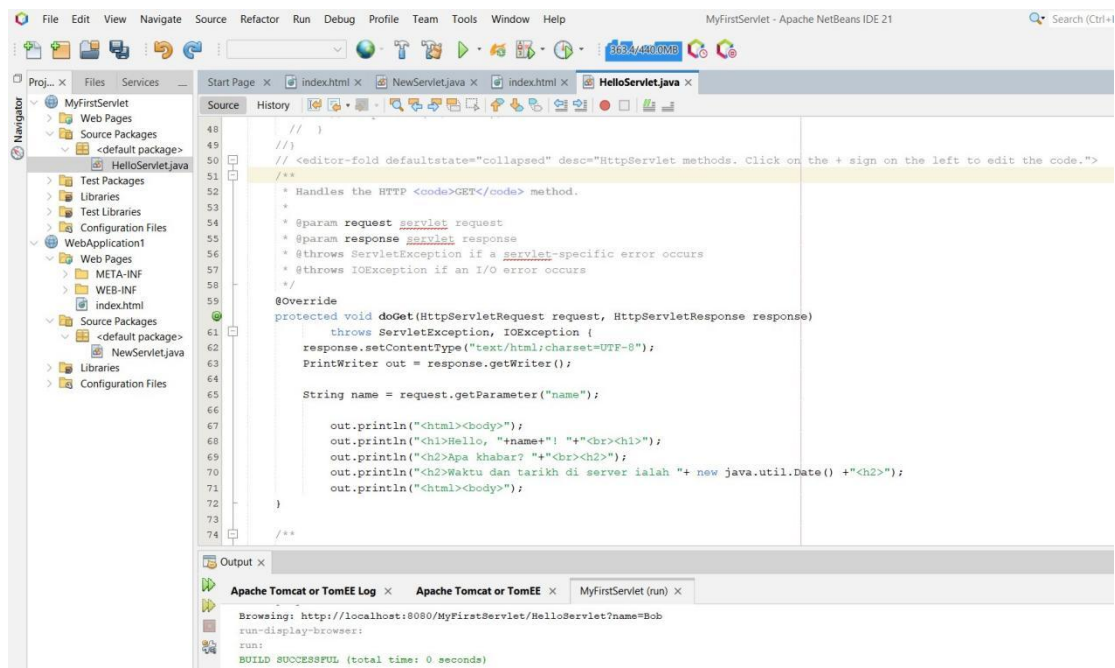
OUTPUT:

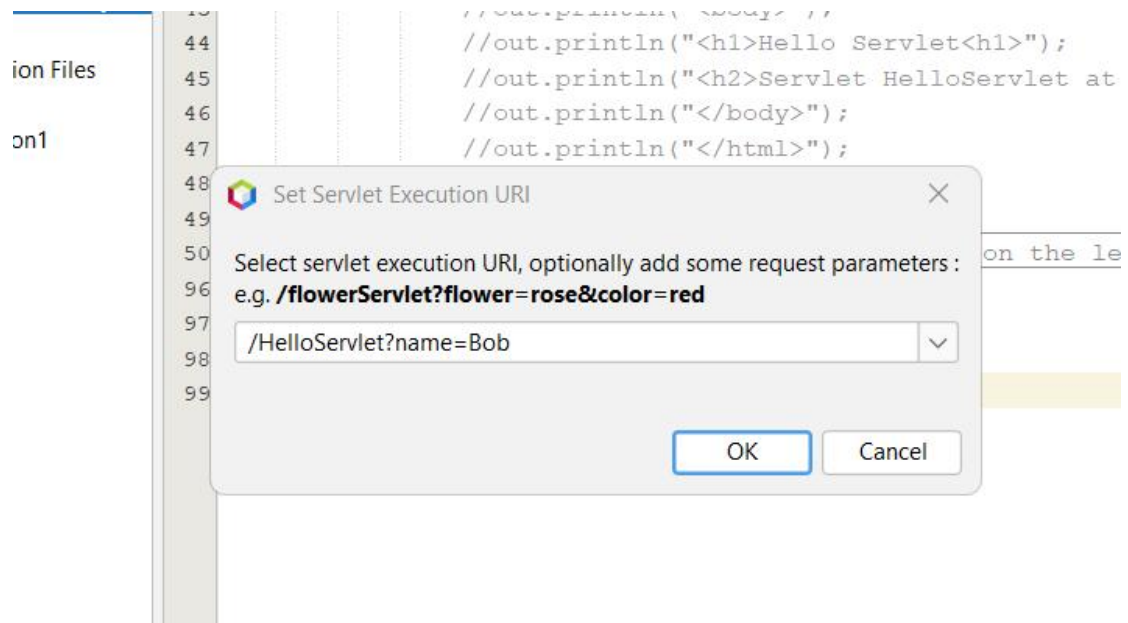


2. Modify the doGet() method:

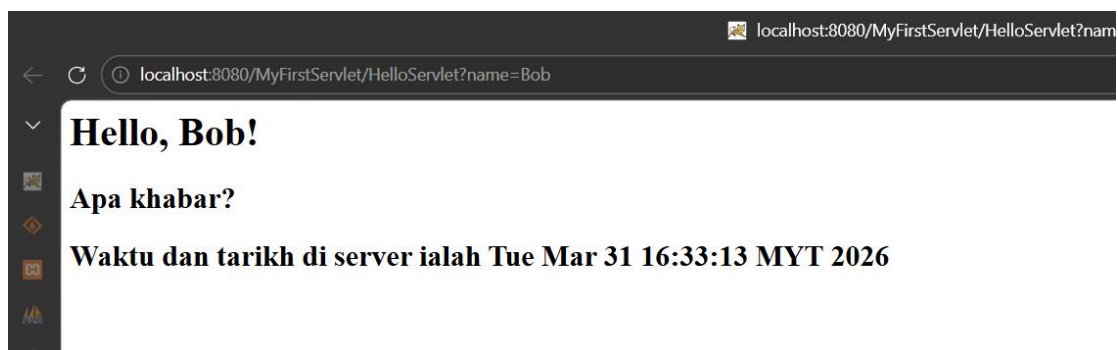
- Locate the doGet() method in HelloServlet.java
- Replace its content with the provided code
- Remove any call to processRequest() inside doGet()

ANSWER:





OUTPUT:

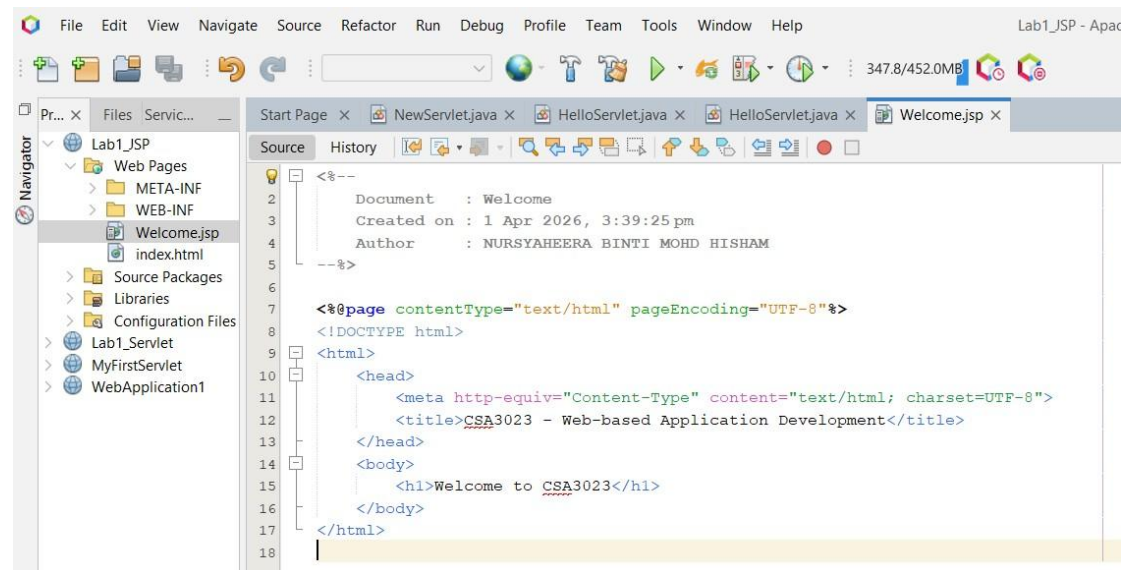


TASK 7: WRITING A SIMPLE JSP PROGRAM

Objective: Create and execute a simple JavaServer Pages (JSP) application using Apache NetBeans and Apache Tomcat.

1. Ensure a successful compilation message appears at the bottom of the NetBeans IDE.

ANSWER:



The screenshot shows an IDE with several tabs: Start Page, NewServlet.java, HelloServlet.java, HelloServlet.java, and Welcome.jsp. The Welcome.jsp tab is active, displaying the following code:

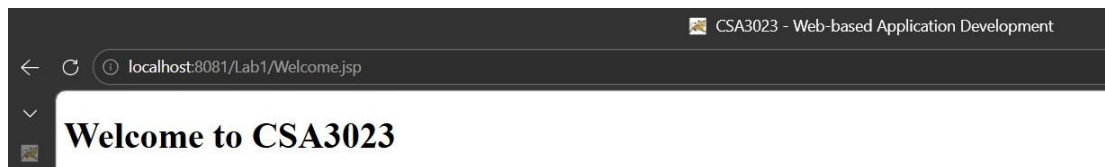
```
<!--
Document : Welcome
Created on : 1 Apr 2026, 3:39:25 pm
Author : NURSYAHEERA BINTI MOHD HISHAM
-->

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>CSA3023 - Web-based Application Development</title>
</head>
<body>
<h1>Welcome to CSA3023</h1>
</body>
</html>
```

The Output window shows the following log messages:

```
Apache Tomcat or TomEE Log x Apache Tomcat or TomEE x Lab1_JSP (clean,dist) x
Created dir: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\web\WEB-INF\classes
Created dir: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\web\META-INF
Copying 1 file to C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\web\META-INF
Copying 3 files to C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\web
library-inclusion-in-archive:
library-inclusion-in-manifest:
Created dir: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\empty
Created dir: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\build\generated-sources\ap-source-output
compile:
compile-jsp:
Created dir: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\dist
Building jar: C:\Users\User\Documents\LAB1_S76276\Lab1_JSP\Lab1_JSP\dist\Lab1_JSP.war
do-dist:
dist:
BUILD SUCCESSFUL (total time: 0 seconds)
```

OUTPUT:



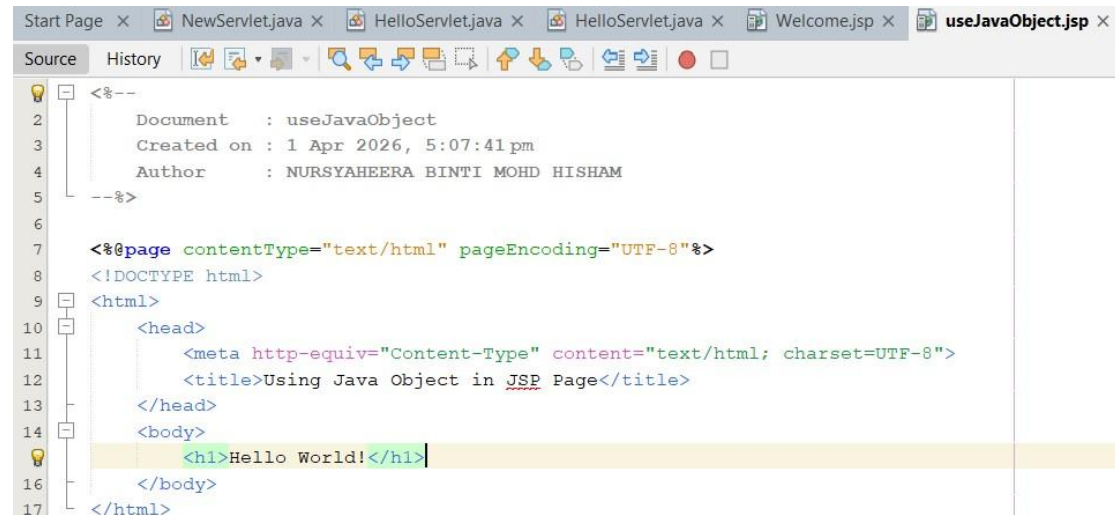
TASK 8: USE JAVA REFERENCE DATATYPE/CLASS WRAPPER IN JSP

Objective: Use Java classes within JSP to display dynamic content such as the current date and implement auto-refresh functionality.

1. Modify the page title:

Using Java Object in JSP Page

OUTPUT:

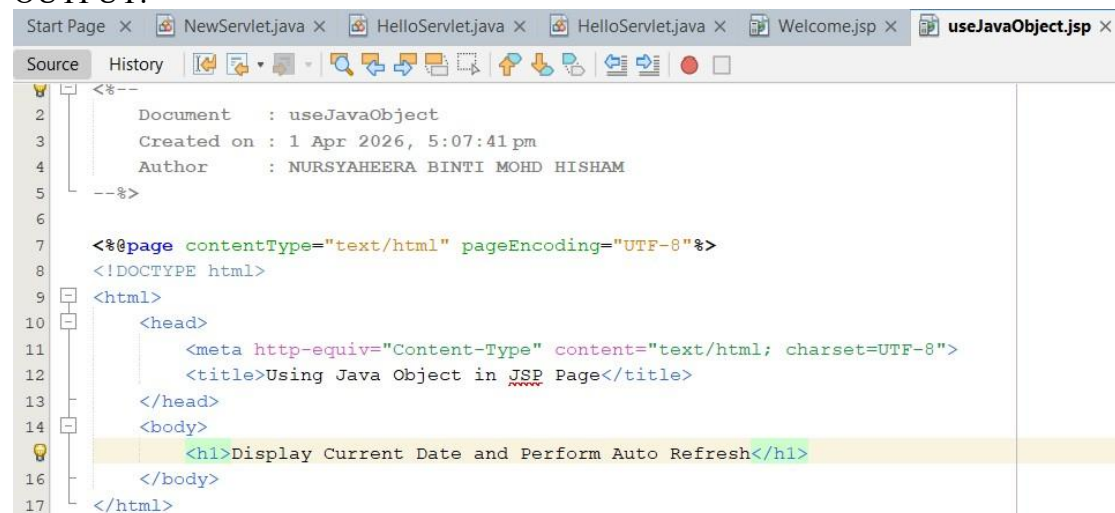


```
1 <%--
2     Document    : useJavaObject
3     Created on  : 1 Apr 2026, 5:07:41 pm
4     Author     : NURSYAHEERA BINTI MOHD HISHAM
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12     <title>Using Java Object in JSP Page</title>
13 </head>
14 <body>
15     <h1>Hello World!</h1>
16 </body>
17 </html>
```

2. Modify the main heading:

Display Current Date and Perform Auto Refresh

OUTPUT:



```
1 <%--
2     Document    : useJavaObject
3     Created on  : 1 Apr 2026, 5:07:41 pm
4     Author     : NURSYAHEERA BINTI MOHD HISHAM
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12     <title>Using Java Object in JSP Page</title>
13 </head>
14 <body>
15     <h1>Display Current Date and Perform Auto Refresh</h1>
16 </body>
17 </html>
```


3. Import the Java utility package for date handling:

```
<%@ page import="java.util.Date" %>
```

OUTPUT:

```
1 <!--
2 Document : useJavaObject
3 Created on : 1 Apr 2026, 5:07:41 pm
4 Author : NURSYAHEERA BINTI MOHD HISHAM
5 -->
6
7 <%@ page import="java.util.Date" %>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>Using Java Object in JSP Page</title>
14 </head>
15 <body>
16 <h1>Display Current Date and Perform Auto Refresh</h1>
17 </body>
18 </html>
19
```

4. Write a JSP scriptlet to create a Date object and display the current date and time:

```
<%
```

```
Date currentDate = new Date();
```

```
out.println("<p>Current date and time is " + currentDate.toString() + "</p>");
```

```
%>
```

OUTPUT:

```
Source History
1 <!--
2 Document : useJavaObject
3 Created on : 1 Apr 2026, 5:07:41 pm
4 Author : NURSYAHEERA BINTI MOHD HISHAM
5 -->
6
7 <%@ page import="java.util.Date" %>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>Using Java Object in JSP Page</title>
14 </head>
15 <body>
16 <h1>Display Current Date and Perform Auto Refresh</h1>
17 <%
18 Date currentDate = new Date();
19 out.println("<p>Current date and time is " + currentDate.toString() + "</p>");
20 %>
21 </body>
22 </html>
```


5. Add code to enable automatic page refresh (e.g., every 5 seconds):

```
<%  
response.setIntHeader("Refresh", 5);  
%>
```

OUTPUT:

```
1  <!--  
2      Document    : useJavaObject  
3      Created on  : 1 Apr 2026, 5:07:41 pm  
4      Author     : NURSYAHEERA BINTI MOHD HISHAM  
5  -->  
6  
7  <%@ page import="java.util.Date" %>  
8  <%@page contentType="text/html" pageEncoding="UTF-8"%>  
9  <!DOCTYPE html>  
10 <html>  
11 <head>  
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">  
13 <title>Using Java Object in JSP Page</title>  
14 </head>  
15 <body>  
16 <h1>Display Current Date and Perform Auto Refresh</h1>  
17 <%  
18     Date currentDate = new Date();  
19     out.println("<p>Current date and time is " + currentDate.toString() + "</p>");  
20 %>  
21  
22 <%  
23     response.setIntHeader("Refresh", 5);  
24 %>  
25 </body>  
26 </html>
```

6. Observe the output in the web browser. The current date and time should be displayed and refreshed automatically

Display Current Date and Perform Auto Refresh

Current date and time is Tue Mar 31 10:46:53 MYT 2026

1.12 EXERCISE

Objective: Apply knowledge of Servlets and JSP to build a simple dynamic web application.

Task: Simple Greeting Web Application

You are required to develop a simple web application that integrates both **Servlet** and **JSP**.

1. Create a new Servlet named GreetingServlet.
2. In the doGet() method:
 - Retrieve a request parameter named name.
 - If the parameter is not provided, set the value to Guest.
 - Forward the request to a JSP page named greeting.jsp.

```
Source History
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
4   */
5
6  import java.io.IOException;
7  import javax.servlet.RequestDispatcher;
8  import java.io.PrintWriter;
9  import javax.servlet.ServletException;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13
14 /**
15  * NAME: NURSYAHEERA BINTI MOHD HISHAM
16  * MATRICS.NO: S76276
17  */
18 */
19 public class GreetingServlet extends HttpServlet {
20
21     /**
22      * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
23      * methods.
24      *
25      * @param request servlet request
26      * @param response servlet response
27      * @throws ServletException if a servlet-specific error occurs
28
29      * @throws IOException if an I/O error occurs
30      */
31     @Override
32     protected void doGet(HttpServletRequest request, HttpServletResponse response)
33         throws ServletException, IOException {
34
35         // 1. Get parameter from URL
36         String name = request.getParameter("name");
37
38         // 2. If null or empty - set default "Guest"
39         if (name == null || name.trim().isEmpty()) {
40             name = "Guest";
41         }
42
43         // 3. Send data to JSP
44         request.setAttribute("name", name);
45
46         // 4. Forward to JSP page
47         RequestDispatcher rd = request.getRequestDispatcher("greeting.jsp");
48         rd.forward(request, response);
49     }
}
```

3. Create a JSP file named greeting.jsp.

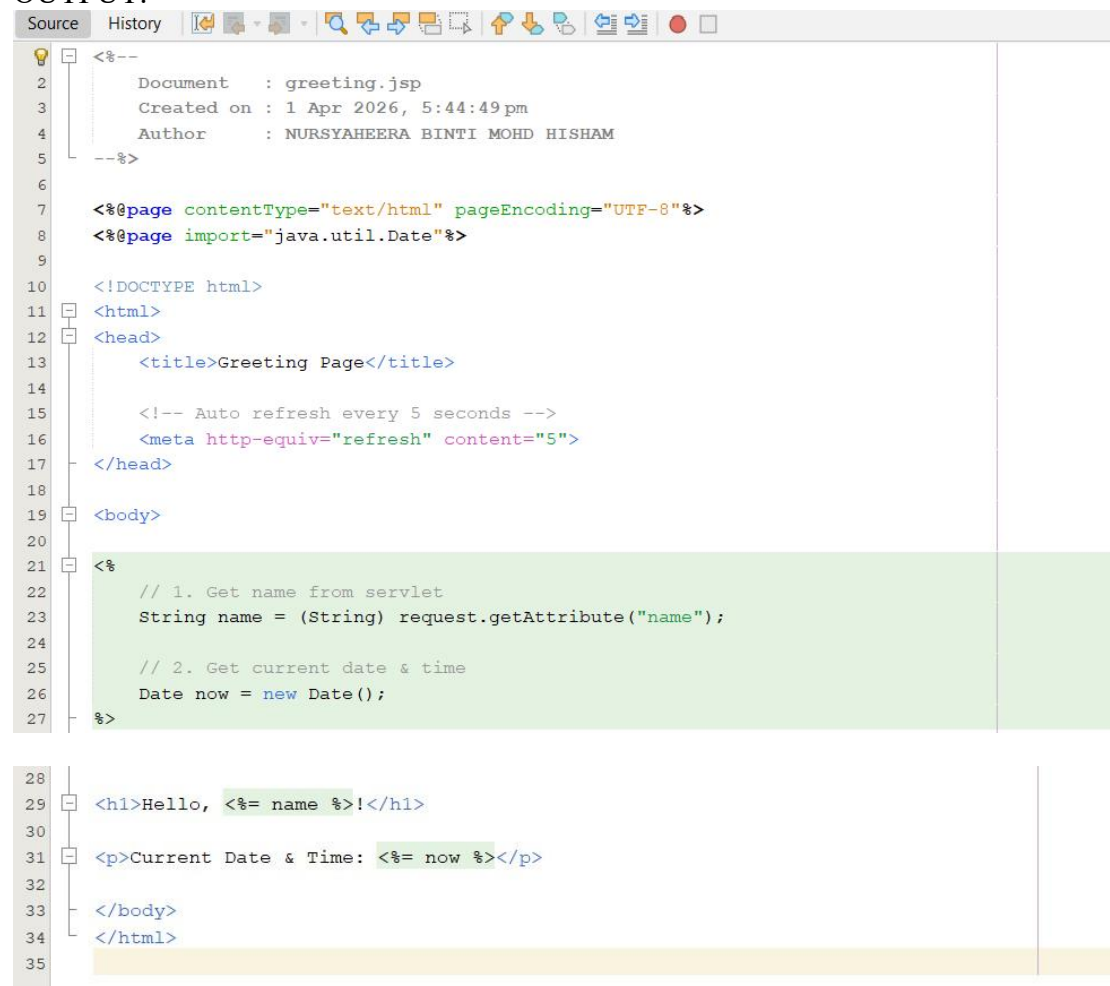
4. In the JSP page:

- Display the message:

Hello, [name]!

- Display the current date and time.
- Ensure the page refreshes automatically every 5 seconds.

OUTPUT:



```
Source History
1 <%--
2     Document    : greeting.jsp
3     Created on  : 1 Apr 2026, 5:44:49 pm
4     Author     : NURSYAHEERA BINTI MOHD HISHAM
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@page import="java.util.Date"%>
9
10 <!DOCTYPE html>
11 <html>
12 <head>
13     <title>Greeting Page</title>
14
15     <!-- Auto refresh every 5 seconds -->
16     <meta http-equiv="refresh" content="5">
17 </head>
18
19 <body>
20
21 <%
22     // 1. Get name from servlet
23     String name = (String) request.getAttribute("name");
24
25     // 2. Get current date & time
26     Date now = new Date();
27 %>
28
29 <h1>Hello, <%= name %>!</h1>
30
31 <p>Current Date & Time: <%= now %></p>
32
33 </body>
34 </html>
35
```

5. Run the application using the following format:

<http://localhost:8080/Lab1/GreetingServlet?name=Ali>

6. Observe the output:

- The name should be displayed correctly.
- If no name is provided, Guest should be shown.
- The date and time should update automatically

OUTPUT:

Hello, Guest!

Current Date & Time: Wed Apr 01 17:56:33 MYT 2026